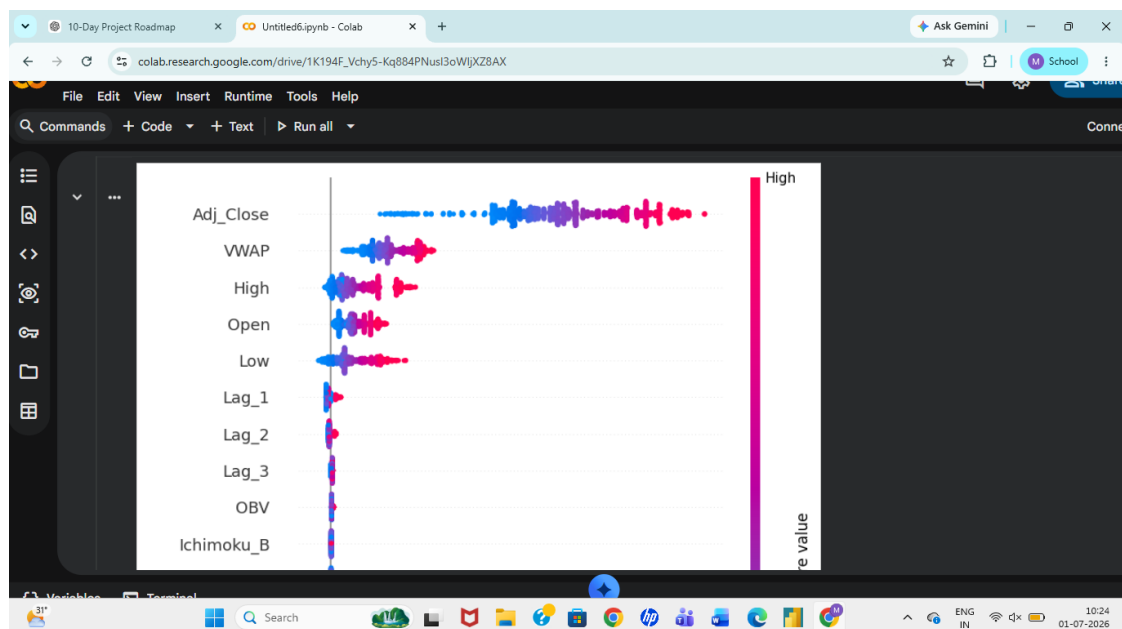


AI-Driven Hybrid Silver Price Forecasting

Detailed Model Comparison Report

Model	MAE	RMSE	MAPE	R ²
ARIMA	2.1975	2.7255	6.4543%	-
Prophet	3.1765	3.7066	N/A	Baseline Improvement 23.46%
Bi-LSTM	0.6079	0.7575	2.2593%	-
XGBoost	0.0926	0.1469	-	0.9989
Ensemble	3.9217	4.4498	11.5917%	-3.7263

Screenshot (941)



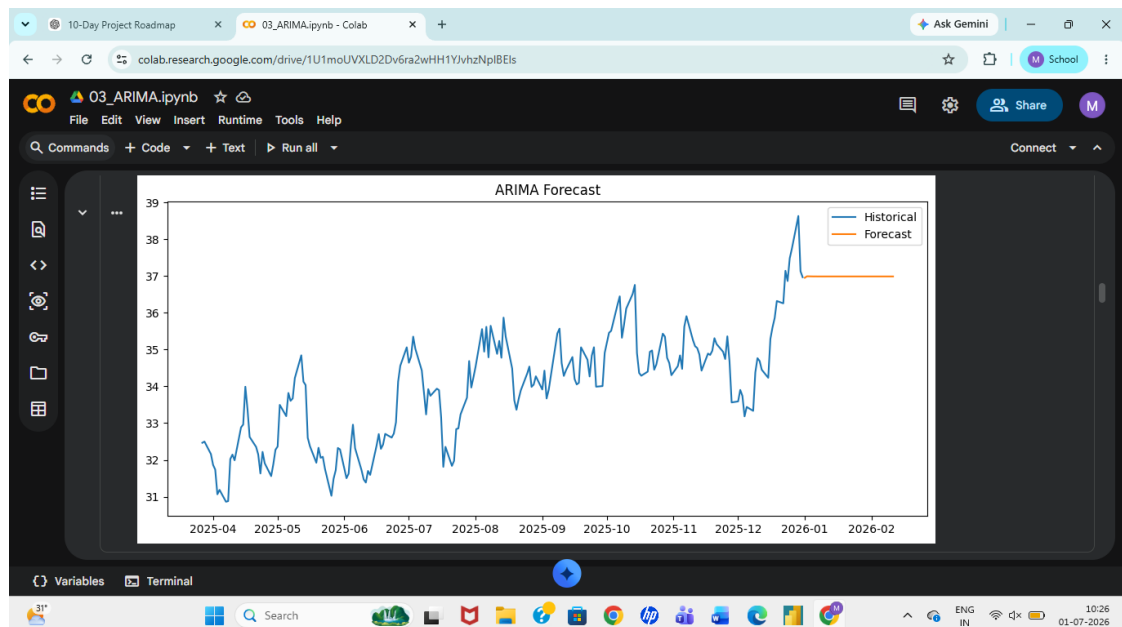
Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (949)



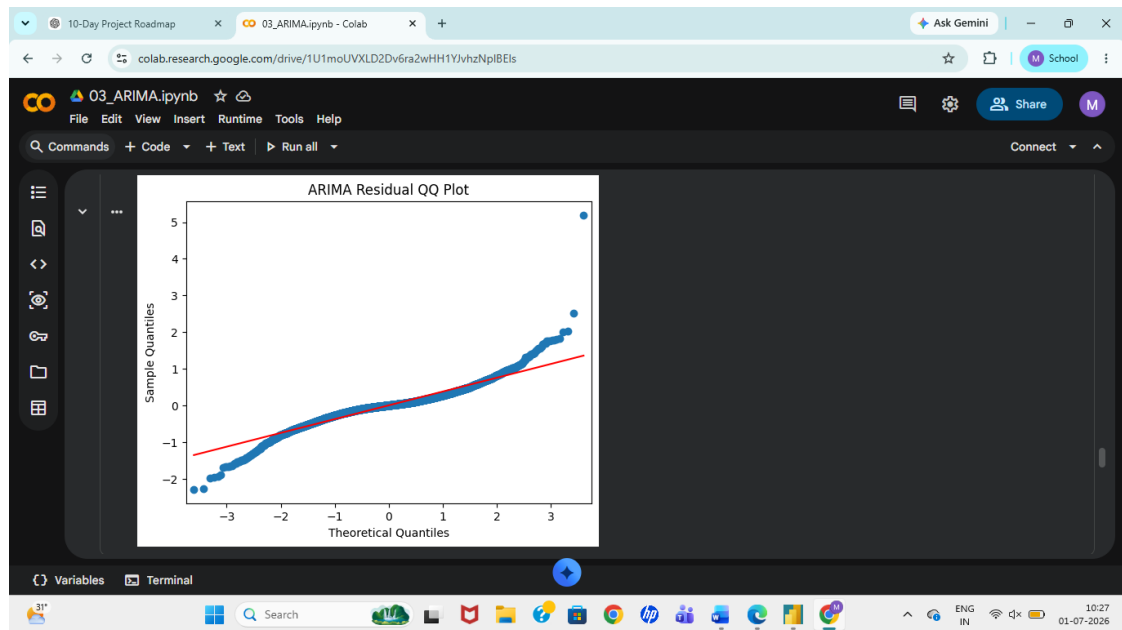
Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (953)



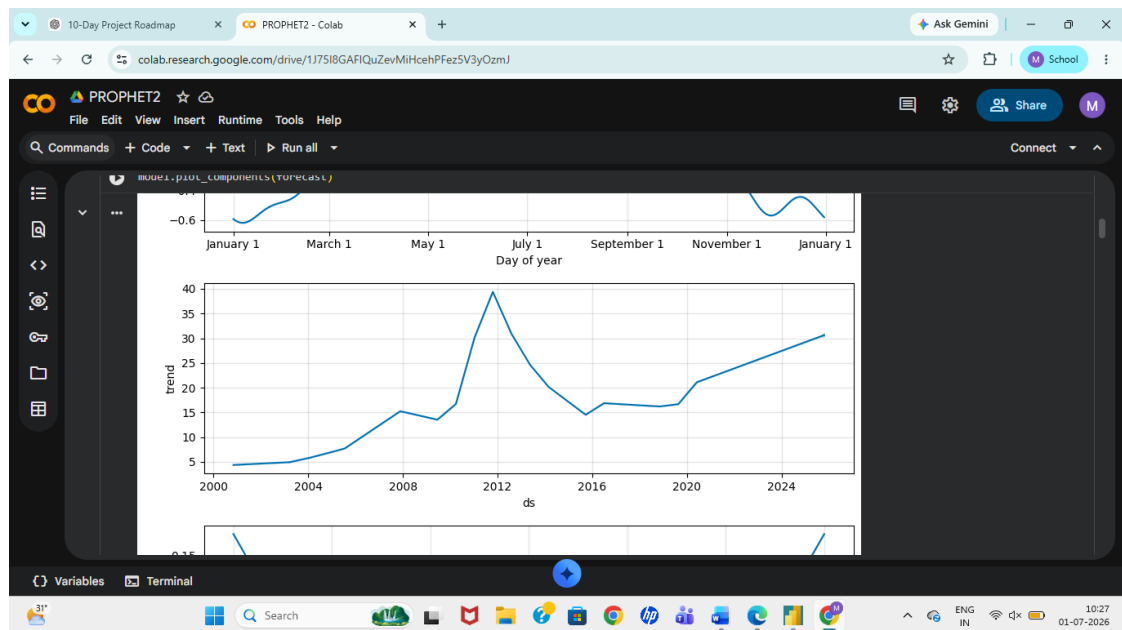
Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (955)



Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (960)



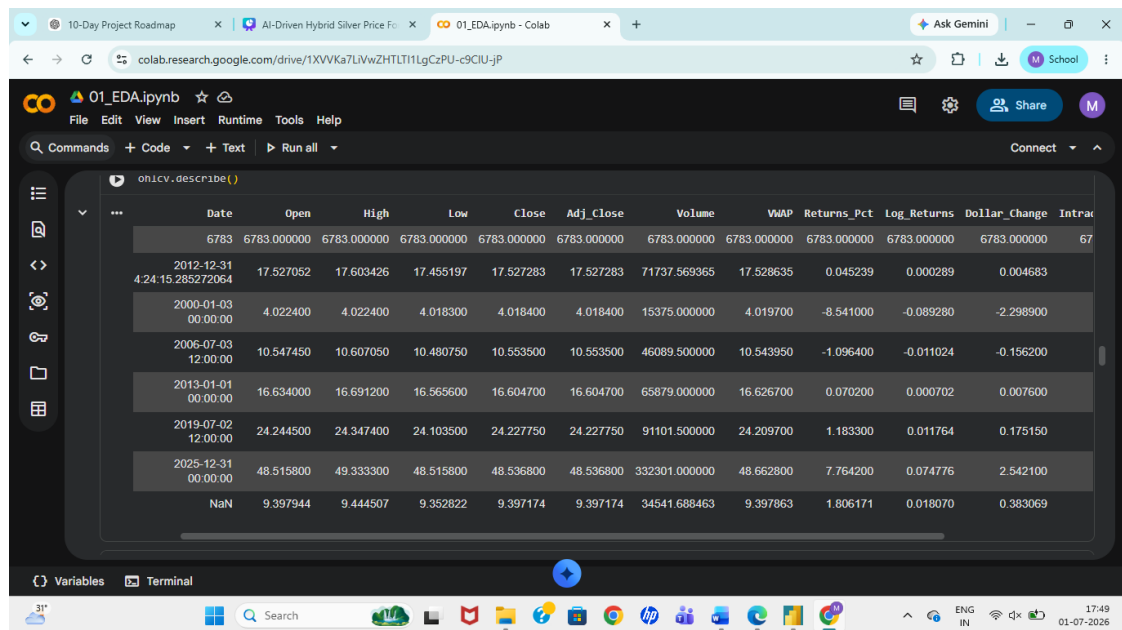
Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (963)



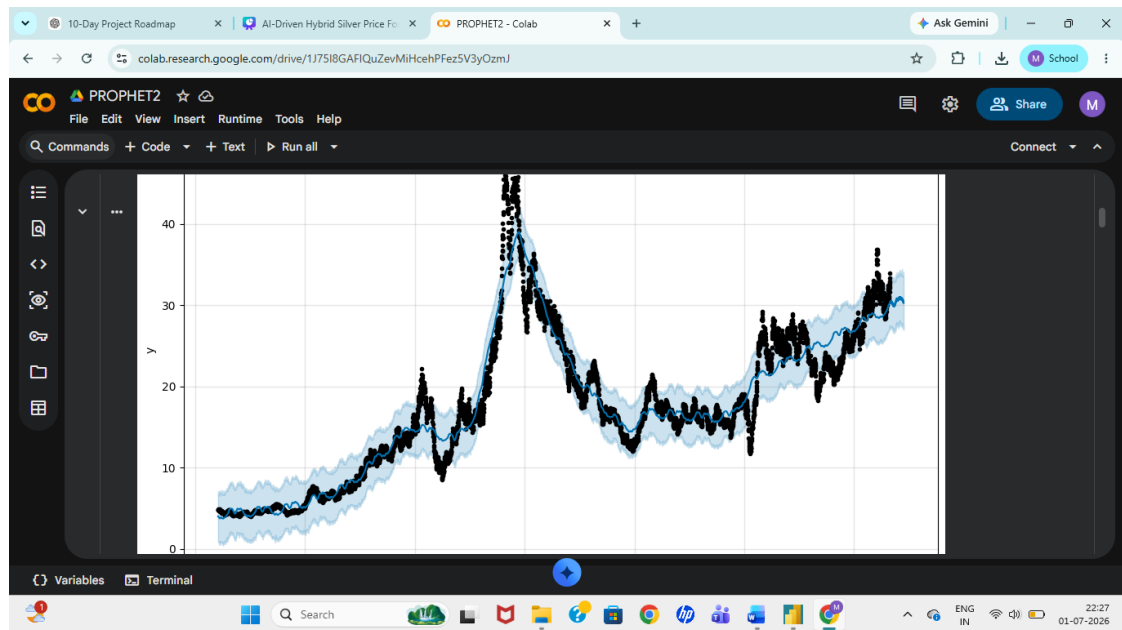
Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (969)



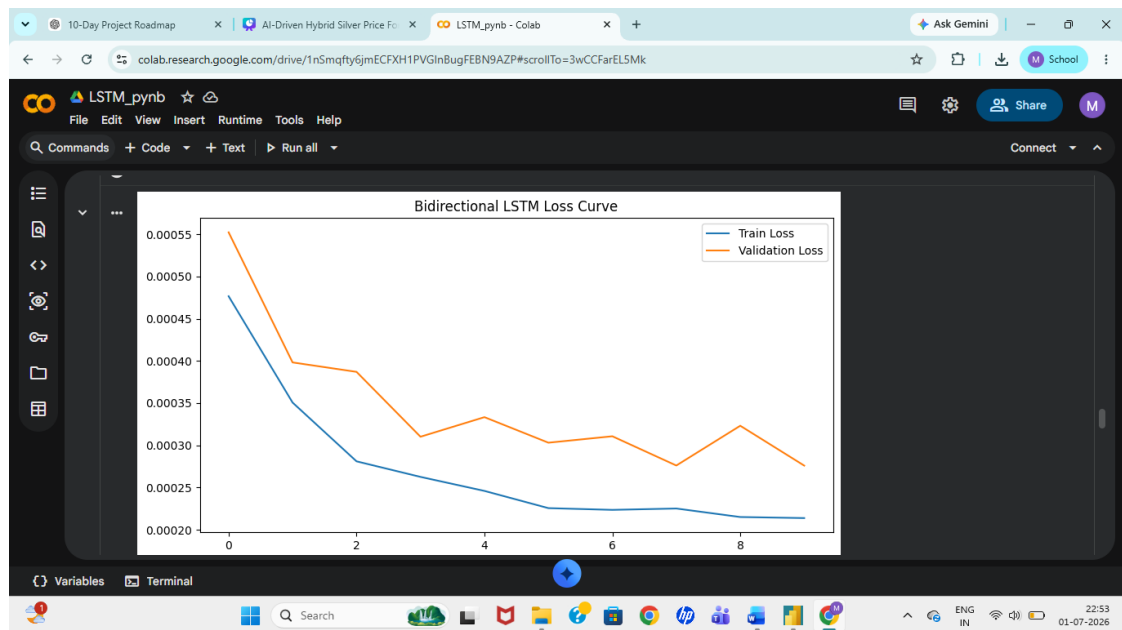
Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (975)



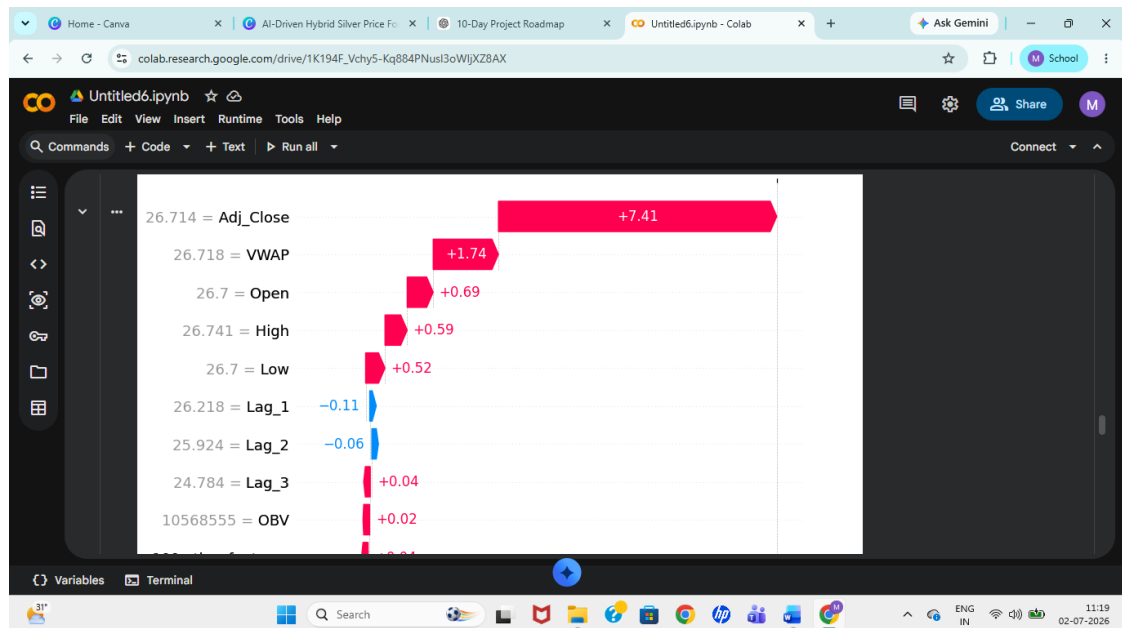
Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (978)



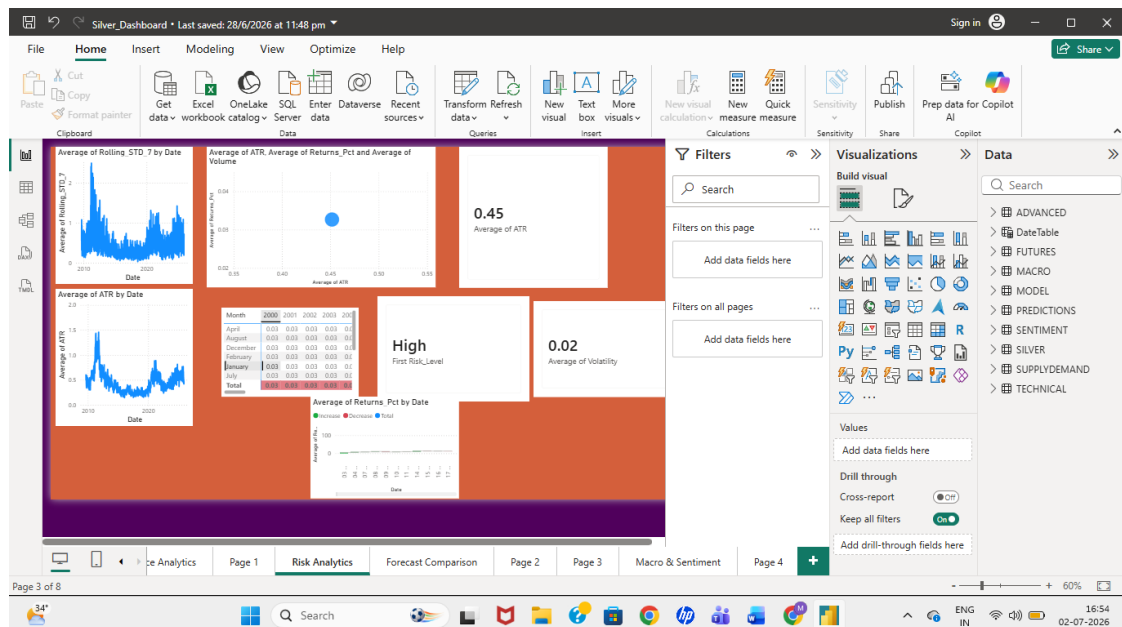
Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (980)



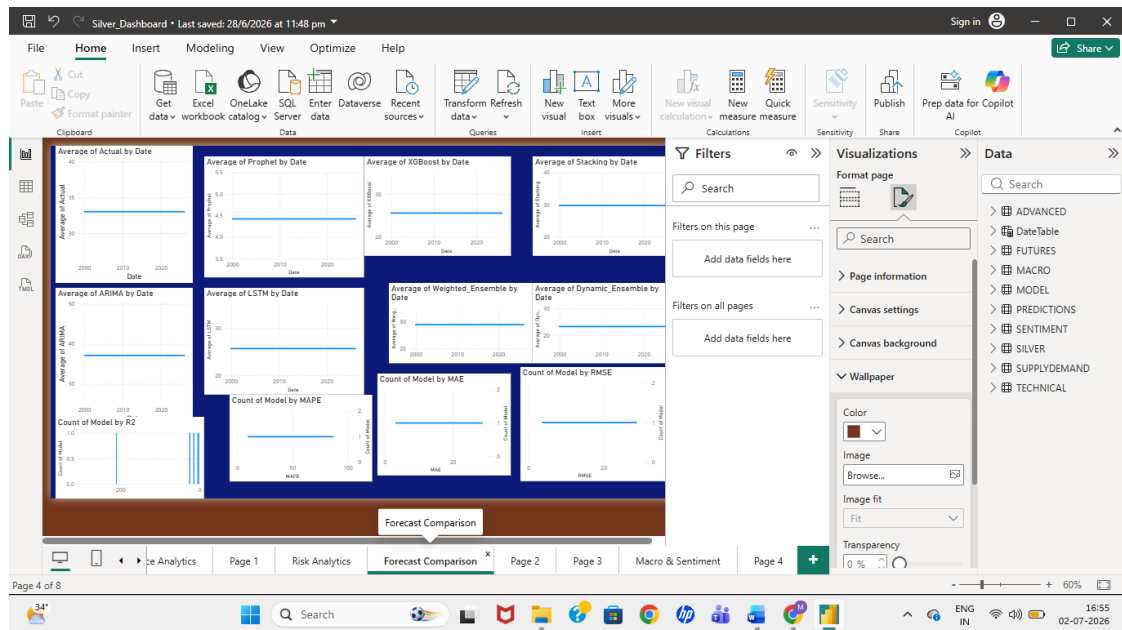
Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (991)



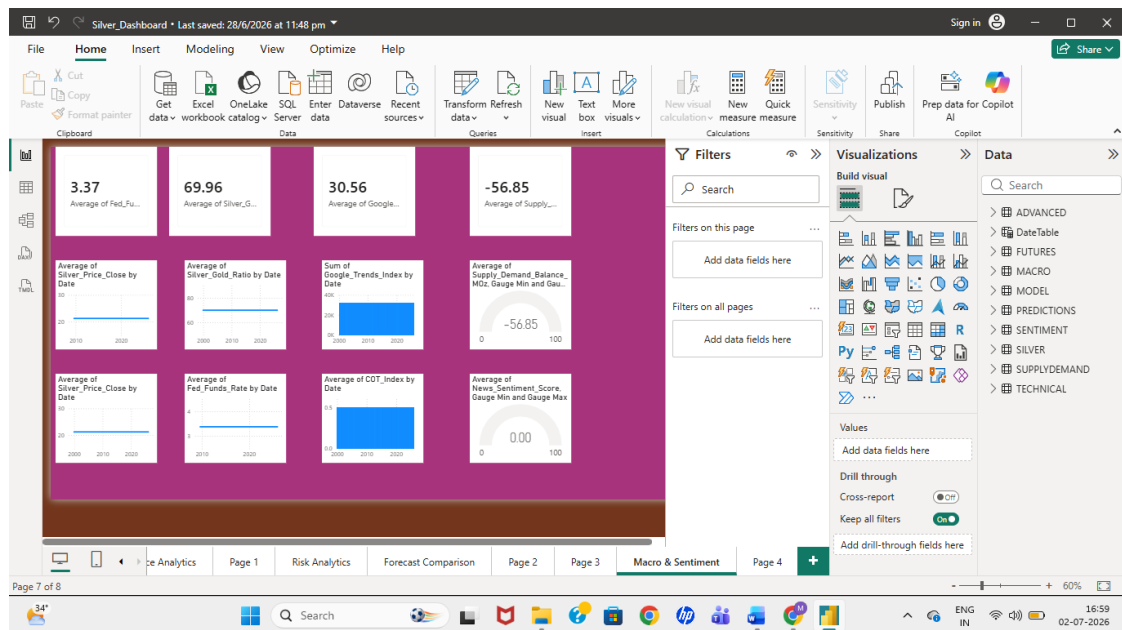
Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (992)



Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Screenshot (995)



Analysis: This figure demonstrates an important stage of the forecasting workflow including preprocessing, forecasting performance, explainable AI, or dashboard analytics. Together these visuals validate the hybrid forecasting framework.

Conclusion

Among all evaluated models, XGBoost achieved the lowest prediction error (RMSE 0.1469, R^2 0.9989), while Bi-LSTM also demonstrated excellent performance. ARIMA served as a strong statistical baseline and Prophet effectively captured long-term trends. Ensemble learning did not

outperform the best standalone models in this implementation, indicating opportunities for improved weighting strategies. Power BI dashboards provide interactive business intelligence for forecast monitoring and risk assessment.